Decoupled Weight Decay Regularization

Department of Computer Science & Artificial Intelligence

Abstract
This paper investigates optimization with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction
Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding Decoupled Weight Decay Regularization.

Section topic: Optimization
Question context: How does the AdamW optimizer differ from the original Adam optimizer regarding weight decay?
Key Finding: AdamW decouples weight decay from the gradient update calculations, applying shrinkage directly to weights rather than treating it as an L2 penalty on gradient moments [Decoupled Weight Decay, p. 4].
Evidence passage: Adam coupled L2 regularization with adaptive gradient moments AdamW decouples weight decay directly from gradient update steps: theta_{t+1} = theta_t - lr * lambda * theta_t - lr * m_hat / (sqrt(v_hat) + eps)

2. Technical Methodology & Empirical Findings

Section topic: Optimization
Question context: How does weight decay act as a regularizer during SGD and Adam training?
Key Finding: Weight decay shrinks parameter magnitudes continuously, penalizing complex over-parameterized representations and promoting generalizability [Decoupled Weight Decay, p. 2].
Evidence passage: Penalizes large weight magnitudes, shrinking parameters toward zero Prevents model capacity from overfitting to spurious sample noise

3. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for Decoupled Weight Decay Regularization. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

4. Technical Methodology & Empirical Findings

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